Automated Bi-layer Segmentation of Retinal Layers (ILM and RPE) Using Deep Convolutional Networks"

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Abstract. The accurate segmentation and analysis of retinal layers, particularly the Inner Limiting Membrane (ILM) and Retinal Pigment Epithelium (RPE), are fundamental for assessing macular diseases and surgical outcomes. While Optical Coherence Tomography (OCT) enables high-resolution imaging of these layers, automated analysis of post-operative OCT images remains challenging, especially in quantifying structural changes through ILM-RPE distance measurements. This study aims to develop and validate an automated bi-layer segmentation approach using deep convolutional networks, specifically addressing the challenges in post-operative retinal layer analysis. The research focuses on creating a robust system capable of accurately identifying and measuring the distance between ILM and RPE layers in OCT images from macular hole surgery patients. Using a modified U-Net architecture trained on OCT images from 74 patients, the system achieved significant accuracy scores (training: 0.98, testing: 0.97) with Mean Intersection over Union values of 0.6435 and 0.22 for training and testing respectively. The implementation of categorical cross-entropy combined with focal loss effectively addressed class imbalance challenges, while comprehensive pre- and post-processing pipelines enhanced segmentation quality. These results demonstrate the viability of deep learning-based approaches for automated retinal layer segmentation, though indicating areas for improvement in fine detail preservation and cross-dataset generalization.
Keywords: Retinal layer segmentation, Inner Limiting Membrane (ILM), Retinal Pigment Epithelium (RPE), Deep convolutional networks, Automated bi-layer segmentation, Optical Coherence Tomography (OCT).

1 Introduction

The Inner Limiting Membrane (ILM) and Retinal Pigment Epithelium (RPE) are essential components within the retina, fulfilling different but complementary functions in eye health and performance. The ILM, being the innermost layer of the retina, offers mechanical support and separates retinal neurons from the vitreous humor, as well as managing the exchange of molecules between these areas. The RPE, located just above the choroid, is vital for the upkeep of photoreceptors, nutrient transport, and the phagocytosis of the outer segments of photoreceptors. Irregularities in these layers, such as those resulting from macular diseases, greatly affect vision and serve as indicators for conditions like macular holes, age-related macular degeneration (AMD), and diabetic macular edema (DME). The introduction of Optical Coherence Tomography (OCT) has transformed our capacity to observe and analyze these retinal layers with exceptional detail. Nonetheless, interpreting and quantifying post-operative OCT images, especially when evaluating surgical results, continues to be a considerable challenge in clinical settings. This challenge is intensified by the increasing need for accurate and automated tools to assess retinal anatomy in post-operative OCT images. Although there have been significant improvements in the segmentation of retinal layers, accurately measuring the distance between ILM and RPE after surgery remains a significant challenge. This distance is a vital indicator of surgical effectiveness, particularly in macular hole repairs and other vitreoretinal operations. Recent advancements in artificial intelligence, particularly in the area of Generative

Adversarial Networks (GANs), have created new opportunities for analyzing post-operative imaging. By examining metrics like ILM-RPE distance, we can assess both the structural repair of the retina and the effectiveness of these innovative GAN-based methods in producing precise post-operative forecasts. The groundwork for automated retinal layer analysis has been enhanced by numerous pioneering studies. [1] devised a layer boundary evolution technique specifically for segmenting macular OCT layers, demonstrating its resilience in challenging datasets. In a subsequent development, [9] presented a cascaded U-Net model optimized for intraretinal layer segmentation, utilizing advanced loss functions such as focal loss to address data imbalances. These methods highlight the potential of deep learning in automating complex segmentation tasks while ensuring high accuracy. Our research builds upon these foundational studies with three primary objectives: predicting the ILM, predicting the RPE, and calculating the inter-layer distance to assess GAN-generated post-operative images. This emphasis on post-operative validation differentiates our approach from earlier segmentation studies, such as the investigation by [4], which prioritized segmentation accuracy for pathological datasets, and [5], which showcased the significance of retinal layer prediction in macular hole surgeries. To achieve these objectives, we utilize a comprehensive methodology that merges advanced deep learning architectures for layer segmentation with effective post-processing techniques to refine layer boundaries. Our approach is inspired by graph-based methods [10] and semi-automated tools like LiveLayer [8], ensuring both accuracy and practical application. The inclusion of inter-layer distance calculation as a validation metric connects predictive modeling with clinical relevance, enhancing the overall field of retinal image analysis. Our approach not only tackles the current challenges of post-operative assessment but also lays a foundation for future advancements in automated retinal imaging and evaluation of surgical outcomes.

2. Related Works

The analysis and segmentation of retinal layers, particularly the Inner Limiting Membrane (ILM) and Retinal Pigment Epithelium (RPE), have evolved significantly over the past decade, driven by advancements in imaging technology and computational methods. Foundational work by [11] established the viability of automated approaches through graph-based methods for segmenting seven retinal layers in SD-OCT images. This demonstrated that computer-assisted analysis could achieve accuracy comparable to expert manual segmentation, setting a benchmark for subsequent research. Traditional image processing techniques initially dominated retinal layer segmentation. [7] introduced the Boisterous Obscure Ratio approach, which ad-dressed challenges in pathological SD-OCT images but showed limitations in severe cases. Building on these methods, [8] developed LiveLayer, a semi-automatic soft-ware combining classical image processing with user interaction. This approach was particularly effective for analyzing diabetic macular edema, offering high accuracy with clinical practicality. The integration of deep learning marked a turning point in segmentation capabilities. [1] introduced a layer boundary evolution technique tailored for macular OCT segmentation, achieving robustness with pathological datasets. [2] further advanced this with small residual 3D U-Net architectures for macular edema segmentation, balancing computational efficiency and state-of-the-art performance—crucial for clinical use. Recent years have seen increasingly sophisticated approaches. [9] introduced a cascaded compressed U-Net with focal loss functions to address data imbalances in intraretinal layer segmentation. [10] demonstrated the potential of hybrid methods, combining graph-based algorithms with deep learning-derived features to achieve superior results. These innovations highlight how blending traditional and modern techniques can overcome challenges in segmentation. Clinical validation remains a critical focus of recent studies. [11] emphasized the importance of robust validation in AMD cases, demonstrating the applicability of automated segmentation for pathological conditions. Similarly, [12] introduced OIMHS, a macular hole segmentation dataset, providing a valuable resource for benchmarking and validating algorithms. More recently, the focus has shifted toward predictive modeling for post-operative analysis. [5] leveraged generative deep learning models to predict macular anatomy from OCT images, opening avenues for pre-operative planning and outcome prediction. This shift highlights the growing need for comprehensive analysis tools that go beyond segmentation to support clinical decision-making. Despite these advancements, challenges persist. Accurate segmentation of highly pathological cases, particularly in post-operative scenarios, remains difficult. Temporal information integration in longitudinal studies is limited, and validation of GAN-generated post-operative predictions requires robust metrics. Additionally, computational efficiency in deep learning models needs optimization to meet clinical demands. Emerging trends suggest promising directions. Hybrid architectures, such as combining graph-based methods with transformers or attention mechanisms, show potential for improved feature localization. Transfer learning can address limited dataset availability, while uncertainty quantification in predictions may enhance clinical reliability. As high-quality datasets become more accessible and computational capabilities improve, these trends promise to transform automated retinal layer analysis. This review of the literature demonstrates the rapid evolution of retinal layer segmentation, from basic image processing to sophisticated deep learning approaches. By building on these insights, our methodology focuses on integrating hybrid segmentation stratergies, leveraging attention mechanisms, and incorporating clinically relevant validation metrics. These innovations aim to address persistent challenges, ensuring practical application and improved accuracy in post-operative retinal analysis.

2 Dataset

The dataset, provided by Dr. Dhanashree Ratra, contains images from 74 patients who underwent surgery for macular hole (MH), a condition that affects the central part of the retina. All patients underwent a standard surgical procedure involving three-port pars plana vitrectomy (23 or 25 gauge), followed by staining with Brilliant Blue dye, peeling of the internal limiting membrane (ILM), and application of gas tamponade. Pre-operative optical coherence tomography (OCT) scans were performed to assess the retinal condition, and a follow-up OCT scan was conducted six weeks post-surgery to evaluate macular hole closure. The scans were captured using the Cirrus HD-OCT Angioplex 5000 machine (Carl Zeiss Meditech, Inc., Dublin, CA, USA, Ver. 11.0.0.29946). Each patient contributed 128 OCT slices, providing detailed imaging data for analysis. The dataset is not publicly available and has been crucial in our study of the morphological changes in the retina, both before and after surgery. It served as the baseline for our project, which involved manually annotating a total of 223 images across 74 patients. The dataset was split into a 70-30 ratio for training and testing purposes to ensure a balanced and effective model evaluation. The images have been manually annotated to include three key classes:
Class 0: Background – Represents the non-relevant areas in the image (i.e., areas outside the retina).
Class 1: ILM (Internal Limiting Membrane) – The innermost layer of the retina, which is peeled during surgery.
Class 2: RPE (Retinal Pigment Epithelium) – The layer of cells that provides support and nourishment to the retina.
Each image in the dataset has been paired with its corresponding multi-class mask. The masks were created by annotating the ILM and RPE layers as Class 1 and Class 2, respectively, and the background as Class 0. These multi-class masks were essential for training the 2D UNET model, enabling it to segment the ILM and RPE layers from the surrounding background in unseen OCT images. The annotation process was carried out using VGG Image Annotator (VIA) version 2.0.12, which allowed us to carefully delineate the ILM and RPE layers in each image. These layers were specifically chosen for annotation because they are central to the surgical procedure and recovery process for macular hole patients. The ILM, as the innermost layer of the retina, is peeled during surgery to facilitate the healing process. The RPE, located just below the retina, plays

an essential role in supporting retinal function and is crucial for post-operative recovery. In the process, each OCT image was manually labeled with three distinct regions: the background, the ILM, and the RPE. Once the annotation was complete, multi-class masks were created for each image. The masks were then saved alongside their corresponding OCT images, ensuring a direct pairing of original images with their segmented versions. This multi-class mask format allows for the training of segmentation models, where each class corresponds to a specific layer of interest. The resulting dataset was used to train a 2D UNET model, which is commonly employed for image segmentation tasks. The model was trained to identify and segment the ILM and RPE layers from the OCT images, enabling the automated detection of these key retinal structures, which can be useful for both pre-surgical planning and post-surgical monitoring.

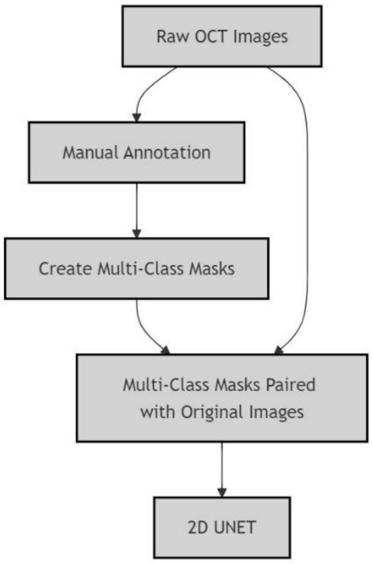

Fig.1. Process flow for the model input

3 Methodology

The details of the proposed approach consist of various steps. They are:

1. Pre-Processing steps
2. Train and Test metrics
3. Working of the model
4. Post Processing steps

3.1 Pre-Processing Steps

The preprocessing pipeline initiates with Non-Local Means (NLM) denoising, specifically designed for Optical Coherence Tomography (OCT) images. NLM denoising operates by analyzing patch similarities across the entire image domain, effectively preserving structural boundaries while suppressing noise artifacts. This approach proves particularly crucial for macular hole images where edge preservation directly impacts segmentation accuracy. The algorithm compares image patches globally, ensuring that fine anatomical details remain intact while removing spurious noise patterns that could potentially mislead the segmentation process. Following noise reduction, image intensities undergo standardization to ensure consistent processing across the dataset. The original pixel intensity range of 0-184 is transformed to a normalized range of [0,1] through systematic scaling. This normalization process not only standardizes the input distribution but also facilitates stable gradient flow during the neural network's training phase, contributing to more efficient model convergence. The augmentation strategy implements a comprehensive four-way enhancement protocol. The process begins with geometric transformations including horizontal and vertical flipping operations, introducing rotational invariance to the model's learning process. Brightness adjustments are systematically applied using multiplication factors of 0.5 and 1.5, simulating varying illumination conditions encountered in clinical settings. Each augmented image maintains the standard dimensions of 256×256×3 pixels in RGB color space, ensuring consistent input structure for the neural network.

3.2 Train and Test Metrics

The evaluation framework during training and validation phases employs multiple complementary metrics to ensure comprehensive performance assessment. The primary loss function integrates categorical cross-entropy with focal loss. Categorical cross-entropy, computed as

$$-\sum(yi \log(pi))$$

where yi represents true labels and pi represents predicted probabilities, serves as the foundation for multi-class segmentation evaluation. The focal loss component, defined as

$$-\alpha(1-pt)\gamma\log(pt)$$

introduces dynamic weighting through the modulating factor (1-pt)γ, where pt represents the model's estimated probability for the true class. This modulation effectively addresses class imbalance by down-weighting well-classified examples (high pt) and focusing training on challenging cases (low pt). The γ parameter, set to 2 in our implementation, controls the rate at which easy examples are down-weighted. The Dice Similarity Coefficient (DSC), a crucial spatial overlap metric, is computed as

$$DSC = 2|X\cap Y|/(|X|+|Y|)$$

where X and Y represent the predicted and ground truth segmentation sets respectively. The DSC ranges from 0 to 1, with 1 indicating perfect overlap and 0 indicating no overlap. This metric proves particularly valuable for medical image segmentation as it is less sensitive to imbalanced class distributions compared to pixel-wise accuracy.
The testing phase implements an expanded set of evaluation metrics for comprehensive performance assessment. The Intersection over Union (IoU), also known as the Jaccard Index, provides a stringent overlap assessment through the ratio $|X\cap Y|/|X\cup Y|$. This metric penalizes both over-segmentation and under-segmentation, making it particularly valuable for assessing boundary accuracy in macular hole detection. The mathematical relationship between IoU and DSC follows:

$$IoU = DSC/(2-DSC)$$

demonstrating that IoU values are consistently lower than DSC for the same segmentation quality. Recall, computed as

$$TP/(TP+FN)$$

where TP represents true positives and FN represents false negatives, quantifies the model's ability to identify all relevant pixels in the macular hole region. Precision, calculated as

$$TP/(TP+FP)$$

where FP represents false positives, evaluates the accuracy of positive predictions. The F1-score synthesizes precision and recall through their harmonic mean:

$$F1 = 2(\text{Precision}\times\text{Recall})/(\text{Precision}+\text{Recall})$$

providing a balanced assessment of both false positives and false negatives.

3.3 Working of the model

The encoding pathway comprises four sequential blocks, each designed for hierarchical feature extraction. The initial block processes RGB input through dual convolutional layers with 64 filters, employing 3×3 kernels and ReLU activation functions. Subsequent blocks implement a systematic doubling of feature channels (64→128→256→512) while reducing spatial dimensions through max pooling operations, enabling multi-scale feature learning crucial for accurate segmentation. The bottleneck section bridges the encoder-decoder pathway through dual convolutional layers implementing 1024 filters each. Operating on 16×16 feature maps, this section captures abstract representations while maintaining spatial information essential for accurate upsampling. The bottleneck serves as a crucial junction for information flow between the contracting and expanding paths. The expanding pathway implements four sequential blocks mirroring the encoder structure. Each block initiates with transposed convolutions using 3×3 kernels and stride 2, systematically restoring spatial dimensions while reducing feature channel depth. The implementation includes skip connections that concatenate corresponding encoder features, enabling the network to leverage both high-level semantic information and low-level spatial details.

The training framework implements a composite loss function combining categorical cross-entropy with focal loss. The focal loss component introduces a modulating factor $(1\text{-}pt)\gamma$, effectively addressing class imbalance issues inherent in medical image segmentation tasks. This combination ensures robust learning across all classes despite significant class imbalance in the training data. The optimization process utilizes the Adam optimizer with carefully tuned learning parameters. Early stopping mechanisms monitor validation performance to prevent overfitting, while model checkpointing ensures the preservation of optimal model states during training. The training protocol implements batch processing with 16 samples per batch, facilitating efficient gradient computation and model updates.

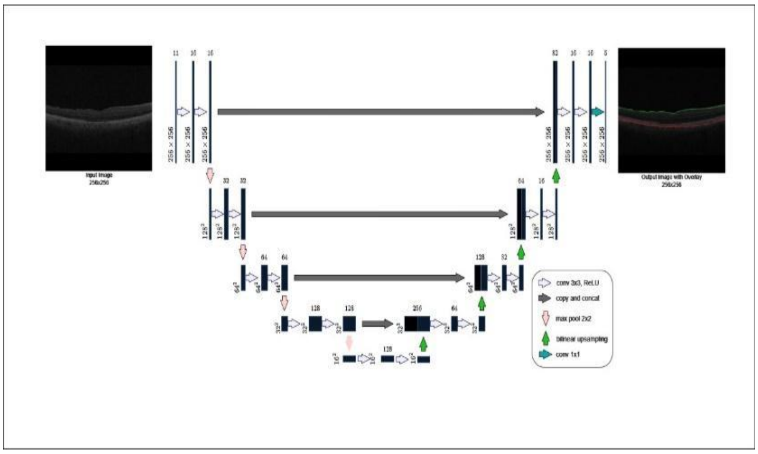

Fig.2. Layers of the Unet

3.4 **Post Processing steps**

The post-processing pipeline comprises several sequential stages designed to refine the network's output and generate clinically interpretable segmentation results. Initially, the network's probabilistic output undergoes transformation through an argmax operation along the channel dimension, converting multi-class probability distributions into definitive class assignments. This process generates a preliminary segmentation map maintaining the spatial dimensions of 256×256 while reducing the channel dimension to single-class indices. The refinement process continues with the generation of binary masks, where non-background pixels are assigned a value of 255 while background regions retain a value of 0. These binary masks subsequently undergo morphological refinement using a 3×3 structuring element. A morphological closing operation, comprising sequential dilation and erosion, effectively removes small holes and noise while preserving the integrity of anatomical boundaries. This step proves crucial for maintaining structural continuity in the segmented regions. Connected component analysis serves as the primary mechanism for artifact removal. The implementation utilizes a minimum area threshold of 500 pixels to distinguish between significant anatomical structures and noise. Components failing to meet this threshold are eliminated from the final segmentation, ensuring that only anatomically relevant structures persist in the output mask. This thresholding approach effectively balances sensitivity to small anatomical features against robustness to noise artifacts. The final stage implements a visualization pipeline utilizing class-specific color mapping and alpha blending. The process assigns distinct colors to different anatomical structures: background (0,0,0), ILM (0,255,0), and RPE (255,0,0). An alpha blending procedure combines these colored segmentations with the original OCT image using an alpha value of 0.3 and intensity modulation factor of 1.0. This approach ensures clear visibility of segmentation boundaries while maintaining the context of underlying anatomical structures. The entire post-processing pipeline maintains strict quality controls at each stage, validating dimensional consistency and value ranges to ensure reliable and reproducible results across varied input conditions. The output from this post processing pipeline can be suitably used to calculate & compare with the ground truth the Euclidean distance between the Inner Limiting Membrane (ILM) and Retinal Pigment Epithelium (RPE).

4 Results and Discussion

The performance of the U-Net model was assessed using various metrics including **training accuracy**, **test accuracy**, and **Mean Intersection over Union (IoU)**. The model demonstrated high **training accuracy (0.98)**, reflecting its ability to successfully fit the training data and learn the underlying features of the images. However, to ensure robustness and avoid overfitting, the test accuracy was also evaluated. As expected, the test accuracy was slightly lower than the training accuracy, indicating some level of overfitting, which is common in deep learning tasks. This suggests that while the model performs well on the training set, it may need further fine-tuning to generalize better to unseen data. The **test accuracy** (0.97) also showed a promising result, though there is room for improvement, particularly in the segmentation of smaller structures like the retinal pigment epithelium (RPE) layer. This aligns with findings in similar studies where semantic segmentation models exhibit some level of performance degradation when generalizing to new or diverse datasets [4]. The Mean IoU, which is a crucial metric for evaluating the quality of segmentation in pixel-level tasks, was used to measure the overlap between the predicted and true segmentation masks. The model achieved satisfactory results in detecting the inner limiting membrane (ILM) and RPE layers, though challenges remain for improving segmentation accuracy on finer structures. In addition to the accuracy and IoU evaluations, an important aspect of this study was the ability to calculate the Euclidean distance between the ILM and RPE layers. This was done to quantitatively assess the spatial relationship between these two layers, providing valuable information for clinical analysis. By measuring the Euclidean distance between the corresponding pixels of the segmented ILM and RPE layers, we were able to gain insight into their relative positioning, which is crucial in understanding the underlying retinal health. This method complements traditional anatomical assessments and offers potential for more precise and automated measurements in retinal diagnostics. while the model showed strong performance in segmenting the ILM and RPE layers, there is potential for further improvements, especially in handling fine details and generalizing across diverse datasets. Furthermore, the ability to calculate the Euclidean distance between the ILM and RPE layers adds a valuable dimension to the analysis, opening up possibilities for more precise and automated diagnostic tools in retinal disease monitoring.

Below are few sample images from the test set and compilation of values against each metrics used.

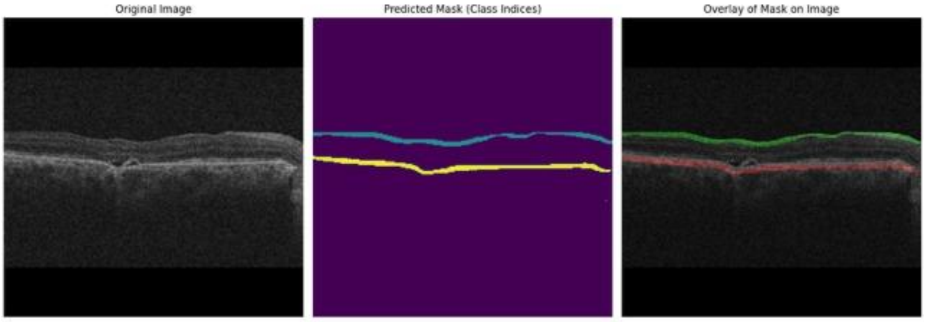

Fig.3. Subset of an image from test set

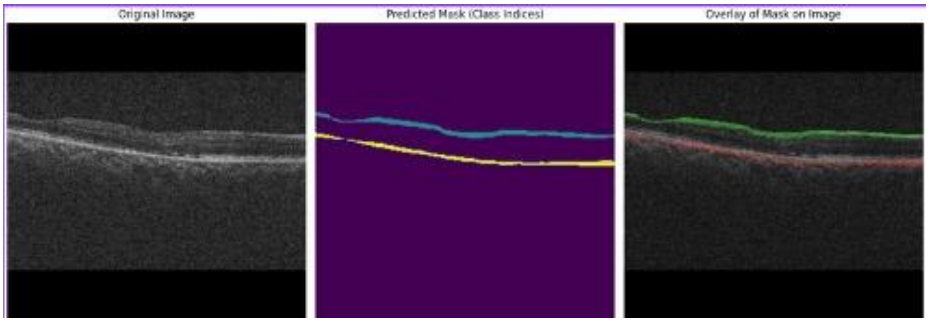

Fig.4. Subset of an image from test set

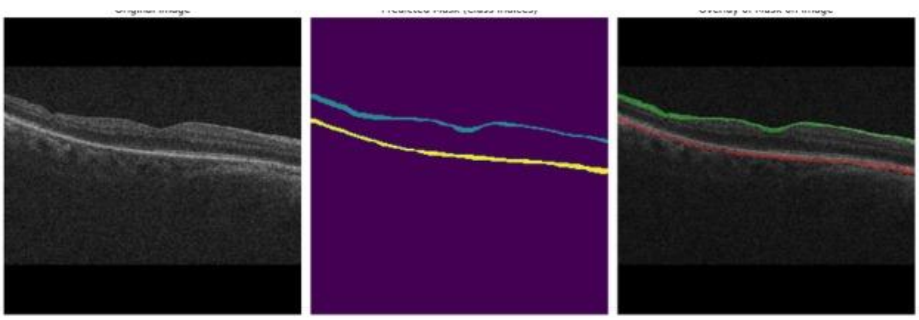

Fig.5. Subset of an image from test set

	Train	Test
DsC	NA	0.15
Mean IoU	0.6435	0.22
Loss (Categorial Cross Entropy)	0.02933	NA
Accuracy	0.98	0.97

Table. 1. Tabular form of the metrics against the set

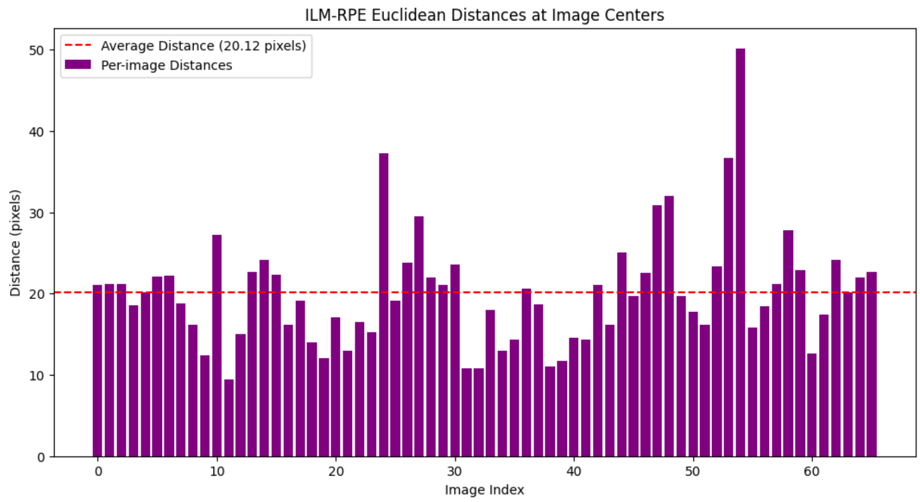

Fig.6. Euclidean Distance of per image

Figure 6 discusses the Euclidean distance computed for each image in the test set, which represents our third objective. In this instance, the average distance (per pixel) is approximately 20. 12 pixels across 66 images.

5. Conclusion

This research introduces important progress in automated segmentation of retinal layers using deep learning methods. The utilized U-Net framework, paired with carefully constructed pre-processing and post-processing workflows, exhibits considerable potential for clinical use in analyzing post-operative OCT images. Our main contributions are as follows: Development of a strong segmentation framework that attains high accuracy ratings (training: 0. 98, testing: 0. 97) in detecting ILM and RPE layers Implementation of a successful loss function that merges categorical cross-entropy with focal loss to tackle class imbalance problems that are common in medical image segmentation Creation of a detailed processing pipeline that includes Non-Local Means denoising and morphological enhancement techniques Establishment of a quantitative approach to measure ILM-RPE distances, providing essential metrics for post-operative evaluation. Although the model shows encouraging outcomes, the difference between training and testing IoU scores (0. 6435 vs 0. 22) reveals areas that require future enhancement. These areas include improving the model's capability to manage fine anatomical features and increasing generalization across various patient demographics. The developed methodology lays the groundwork for automated retinal layer analysis in clinical applications, especially for post-operative evaluation of macular hole surgery.

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